

Neural Survival Models: LongitudinalDeepHit

Part E(iii) — Transformer Encoder for Competing Risks

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Why go beyond Cox? Mortgage prepayment is path-dependent.

Cox / Andersen-Gill limitations

- Proportional hazards: covariate effect is **constant over time**
- Single scalar score — cannot model the **S-curve shape** of prepayment
- Competing risks require separate model fits; no shared representation
- Cannot exploit the **sequence structure** of monthly covariate history

What a sequence model can do

- Read the full monthly path $[x_1, \dots, x_T]$ — **burnout is encoded**
- Produce nonlinear, time-varying hazard rates $\lambda_k(t \mid x)$
- Model prepayment and default **jointly** via shared encoder
- Attention weights identify **which months** drive the prediction

Why this architecture? Three design decisions.

Alternatives rejected

- **LSTM** — vanishing gradients at $T=120$; attends only to adjacent months; hard to interpret
- **Single-context head** — no time index in head; near-constant hazard curve; gradient flows only through last hidden state
- **Two separate models** — violates independent censoring (defaulted loans are riskiest borrowers, not random exits); loses shared macro signal

Why this combination

- **Causal Transformer** — reaches any past month in one attention step; gradient flows through every position; weights are interpretable
- **Per-timestep head** — $\lambda(t)=\text{head}(h_t)$: hazard conditioned on full causal history $x_1 \dots x_t$; every month trained independently
- **Joint encoder** — shared representation captures that the same macro shock raises default and suppresses prepayment; chain product enforces $\text{CIF}_1 + \text{CIF}_2 + S = 1$

LongitudinalDeepHit: Transformer encoder with two CIF output heads.

Architecture

Input	$(B, T_{\max}=120, D=13)$ padded sequences
Proj.	Linear $D \rightarrow d_{\text{model}}=64$
PE	Sinusoidal positional encoding on loan age
Enc.	$\times 2$ pre-norm layers, 4 heads, causal mask
Head 1	FC $64 \rightarrow 64 \rightarrow 1$ per h_t , Sigmoid $\Rightarrow \lambda_1(t)$
Head 2	FC $64 \rightarrow 64 \rightarrow 1$ per h_t , Sigmoid $\Rightarrow \lambda_2(t)$
CIFs	Competing-risk chain product

Competing-risk CIFs

$$\begin{aligned}\text{CIF}_k(t) &= \sum_{s=1}^t \lambda_k(s) \cdot S(s-1) \\ S(t) &= \prod_{s=1}^t (1 - \lambda_1(s))(1 - \lambda_2(s))\end{aligned}$$

Guarantees $\text{CIF}_1 + \text{CIF}_2 + S = 1$ at every t .

Training objective

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \alpha \mathcal{L}_{\text{rank}}, \quad \alpha = 0.2$$

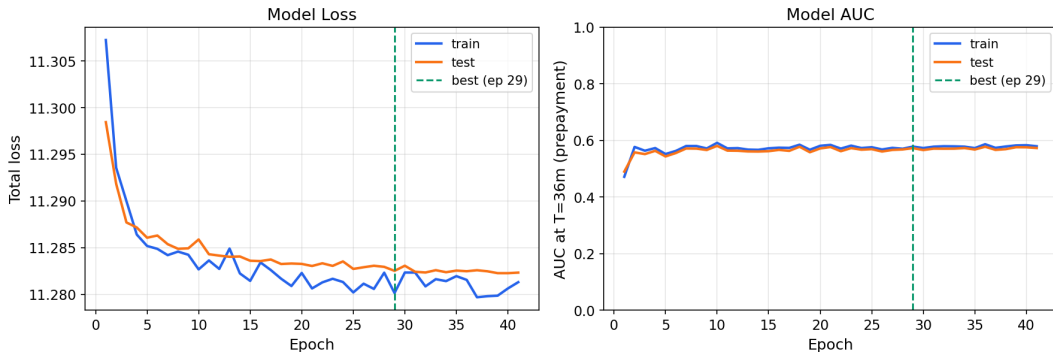
NLL: discrete-time cause-specific likelihood

Ranking: soft pairwise concordance
(DeepHit-style)

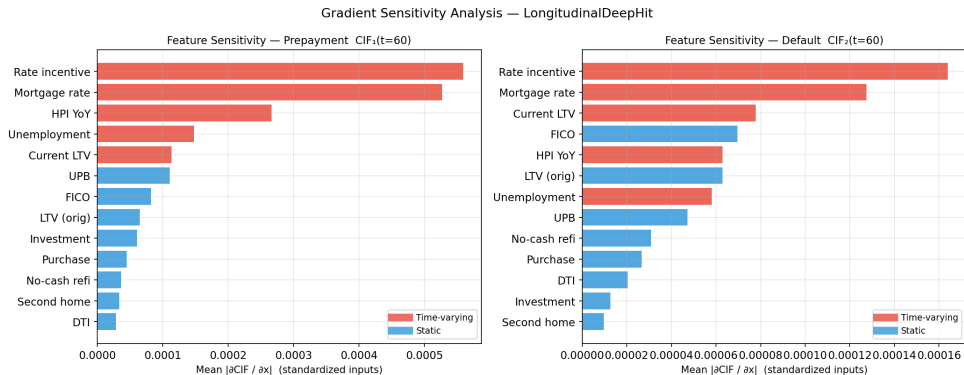
Params	76,290
Data	101K loans, 4.5M rows
Epochs	41 (best ep. 29)
LR	3×10^{-4} , cosine
Loss	11.30 $\rightarrow$ 11.28
Rank	1.97 $\rightarrow$ 1.84

NLL collapses to $\lambda \approx 0$; ranking loss drives AUC improvement.

LongitudinalDeepHit — Training Curves



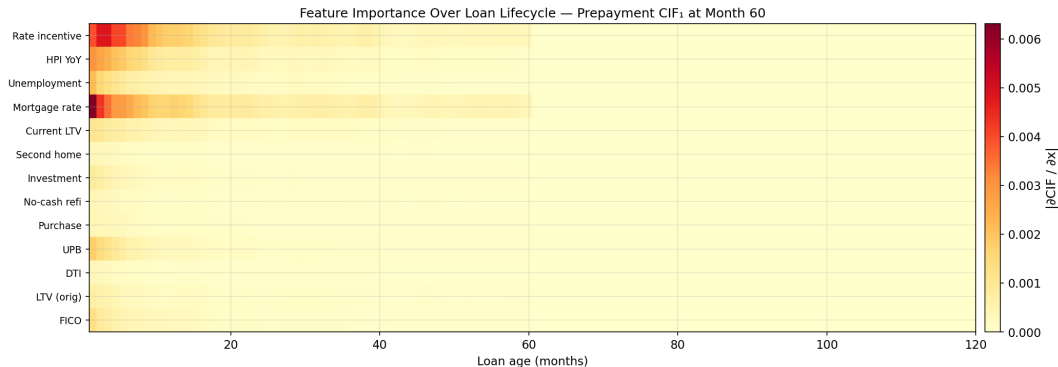
Rate incentive and current LTV dominate prepayment; FICO dominates default.



Red = time-varying features. Blue = static features.

Dynamic features rank above static for prepayment; FICO and unemployment lead for default — consistent with cause-specific Cox in $E(i)$.

Feature importance shifts across the loan lifecycle — unique to sequence models.



rate_incentive dominates early months; Current LTV rises as equity accumulates and gates the refi option.
Mortgage rate is dark throughout — the model learns the incentive *spread* matters, not the rate level.

Default AUC matches DeepCox; prepayment underperforms due to NLL collapse.

Prepayment CIF₁ — AUC at fixed horizons

Model	T=12	T=24	T=36	T=60
Part D DeepCox [†]	0.684	0.701	0.721	0.729
DeepHit (20K) [‡]	0.641	0.622	0.572	0.515
DeepHit (344) [§]	0.677	0.647	0.565	0.334

Default CIF₂ — AUC at fixed horizons

Model	T=12	T=24	T=36	T=60
Part D DeepCox [†]	0.684	0.701	0.721	0.729
DeepHit (20K) [‡]	0.687	0.708	0.707	0.726

[†] 597K canonical [‡] 20K hold-out [§] 344 canonical loans

Default AUC **matches DeepCox** at every horizon with **6× fewer** training loans — temporal covariate history compensates for data volume.

Key Takeaways

1 **Default AUC roughly matches DeepCox**

Using $6\times$ fewer training loans. Temporal covariate history compensates for data volume.

2 **Gradient sensitivity independently replicates Cox findings**

rate_incentive + Current LTV drive prepayment; FICO + unemployment drive default — recovered from data, not imposed.

3 **Feature importance shifts across the loan lifecycle**

rate_incentive dominates early months; Current LTV rises as equity accumulates. This time-resolved view is unique to sequence models.

4 **Known limitation: NLL collapse at 1.6% monthly default rate**

Survival gradient overwhelms event gradient in epoch 1; AUC is preserved but Brier scores are elevated (0.07–0.30). Fix: focal-loss upweighting.

Thank you

Questions?

Lee et al. (2018) *DeepHit* · Sadhwani et al. (2021) *J. Fin. Econometrics* · Vaswani et al. (2017) *Attention Is All You Need*